MILITARY SPECIFICATION

PRIMER, BATTERY, ELECTRIC PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U. S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

10 SCOPE

1.1 This specification covers the parts for, and the loading, assembling and packing for one type of primer designated as Primer, Battery, Electric.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on data of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621	-	Boxes, Wood, Nailed and Lock Corner

MILITARY

MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, packaging and Packing of
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-650	-	Explosives, Sampling, Inspection and Testing

DRAWINGS (see 6.5)

U. S. Army Armament Research and Development Command

PRODUCT DRAWINGS

9292624	-	Primer, Battery, Electric

INSPECTION EQUIPMENT

9275627	-	Test Fixture

2.2 Other publications. Tne following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, “49 CFR 100-199 (latest revision)”).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawing and specifications.

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3.2 Electrical battery primer (battery primer). The battery primer shall comply with all requirements specified on Drawing (Dwg.) 9292624, and associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Moisture content.

3.3.1 Barium nitrate. The moisiture content of the barium nitrate at the time of loading the battery primer shall not exceed 0.50 percent when tested as specified in 4.5.2.

3.3.2 Lead azide. The moisture content of the lead azide at the time of loading the battery primer shall not exceed 0.50 percent when tested as specified in 4.5.3.

3.4 Functioning tests.

3.4.1 Sensitivity and output.

3.4.1.1 No-fire test (70° ± 10°F). The battery primer shall not fire when subjected to a current of 10 milliamperes (ma.) for 10 seconds minimum (min).

3.4.1.2 All-Fire output test (70° ± 10°F). The battery primer shall function on the first attempt and shall indent the specified witness plate for not less than 0.040 inch nor more than 0.120 inch, and shall expand the diameter of the witness sleeve by not less than 0.003 nor more than 0.010 inch when tested as specified in 4.5.4.2.

3.5 Environmental tests.

3.5.1 Insulation resistance between twisted electrodes and cup, thermal shock, vibration and function. The battery primer shall comply with the insulation resistance requirements of Dwg. 9292624 and with the requirements of 3.4.1 after being exposed to thermal shock and vibration as specified in 4.5.5.1.

3.5.2 Waterproofness. The battery primer shall comply with the requirements of 3.4.1.2 after being submerged in water to a minimum (min.) depth of two (2) inches for not less than 48 hours, at a temperature of 70 degrees F plus or minus 10 degrees, when tested as specified in 4.5.5.2.

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3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be clean and free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, oil, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Cup, Battery Primer	9292684	10
Ferrule	9292686	10
Sleeve, Header	9292690	10
Header	9492689	25
Header, Bridged	9292688	25
Bridged Header and Ignition Assembly	9292687	25
Primer, Battery, Electric	9292624	150

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4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63316(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Electric Battery Primer and Components				See Below
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup, Battery Primer (Dwg. 9292684)				
	Examination for defects	10		3.2	4.4.2.1
	Ferrule (Dwg. 9292686)				
	Examination for defects	10		3.2	4.4.2.3
	Sleeve, Header (Dwg. 9292690)				
	Examination for defects	10		3.2	4.4.2.4
	Header (Dwg. 9292689)				
	Examination for defects	25		3.2	4.4.2.5
	Pull Test	25	0-1	3.2	4.5.6
	Header, Bridged (Dwg. 9292688)				
	Examination for defects	25		3.2	4.4.2.6
	Bridged Header and Ignition Assembly (Dwg. 9292687)				
	Examination for defects	25		3.2	4.4.2.7

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Electric Battery Primer and Components				See Below
					NEXT HIGHER ASSEMBLY ----
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Primer, Battery, Electric (Dwg. 9292624)	150			
	Examination for defects	Note 1		3.2	4.4.2.8
	Bridgewire resistance test	Note 1		3.2	4.5.7
	Insulation resistance test	Note 1		3.2	4.5.8
	Sensitivity and Output Tests	75			4.5.4
	No-Fire Test		75-0-1	3.4.1.1	4.5.4.1
	All-Fire output test:				
	Sleeve enlargement		75-0-1	3.4.1.2	4.5.4.2
	Dent in place		75-0-1	3.4.1.2	4.5.4.2
	Environmental Tests				
	Thermal shock, vibration and function	25			
	Insulation resistance (after vibr)	Note 2	25-0-1	3.5.1	4.5.5.1
	No-fire test		25-0-1	3.4.1.1	4.5.5.1
	All-fire test				
	Sleeve enlargement		25-0-1	3.4.1.2	4.5.5.1
	Dent in plate		25-0-1	3.4.1.2	4.5.5.1
	Waterproofness Test	50			
	All-fire test:			3.5.2	4.5.5.2
	Sleeve enlargement		50-0-1	3.4.1.2	4.5.4.2
	Dent in plate		50-0-1	3.4.1.2	4.5.4.2

NOTES: 1. The entire quantity of battery primers required for the tests below shall be subjected to the examinations specified prior to function testing. 2. Sample of 25 units shall be selected from units that previously met insulation resistance test criteria.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of primers shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Lead azide from not more than one lot.

c. Lead styphynate from not more than one lot. Lead styphnate with lacquer (SPOT CHARGE) from one batch or blending of batches which results in homogeneity of all lead styphnate with lacquer used in loading the entire ignition assembly lot.

d. Barium nitrate from not more than one lot.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292684
4.4.2.1	Cup, Battery Primer				NEXT HIGHER ASSEMBLY 9292624
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside Diameter		0.40%	3.2	Gage
102	Outside Diameter		0.40%	3.2	Gage
103	Overall length		0.40%	3.2	Gage
104	Thickness through closed end		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63316(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292691
4.4.2.2	Electrode				NEXT HIGHER ASSEMBLY 9292689
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Diameter of electrode		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63316(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Ferrule				9292686
					NEXT HIGHER ASSEMBLY 9292624
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	True position of O.D. to I.D.		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
104	Inside Diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63316(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Sleeve, Header		9292690
			NEXT HIGHER ASSEMBLY 9292689

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
104	True position of I.D. to O.D.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63316(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292689
4.4.2.5	Header				NEXT HIGHER ASSEMBLY 9292688
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Separation between leads, 2 places		0.40%	3.2	GAGE
102	Pull Test of Electrode, 2 places	(a)	0-1	3.2	4.5.6
103	Solderability of plated electrode, 2 places	(a)	0-1	3.2	Method 208 of MIL-STD-202
104	Meniscus, Maximum (max.)		0.40%	3.2	GAGE
Minor					
201	Evidence of poor workmanship		0.65%	3.7	VISUAL

NOTES:
(a) Sample as per MIL-STD-105, Level S-3.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292688
4.4.2.6	Header, Bridged				NEXT HIGHER ASSEMBLY 9292687
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Bridgewire material improper			3.2	CERTIFICAITON
102	Bridgewire weld missing or inadequate, 2 places		0.40%	3.2	VISUAL - MANUAL*
103	Length of pins from sleeve, Max.		0.40%	3.2	GAGE
Minor					
201	Evidence of poor workmanship		0.65%	3.7	VISUAL

NOTES:
* Using magnifier.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63316(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292687
4.4.2.7	Bridged Header and Ignition Assembly				NEXT HIGHER ASSEMBLY 9292624
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lead styphnate on outside diamter of header		0.40%	3.2	GAGE
102	Height of Spot charge above header, max.		0.40%	3.2	GAGE
103	Radial separation of spot charge from outside diameter of header, min.		0.40%	3.2	GAGE
104	Spot charge is cracked, separted or has voids (after drying)		0.40%	3.2	VISUAL - MAGNIFICATION
Minor					
201	Evidence of poor workmanship		0.65%	3.7	VISUAL

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.8	Primer, Battery, Electric		9292624
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup cracked or split		100%	3.2	VISUAL
2	Explosive on exterior of cup		100%	3.2	VISUAL
Major					
101	Diameter, Max.		0.40%	3.2	GAGE
102	Bare electrodes not shorted (twisted)		100%	3.2	VISUAL
103	Length of body		0.40%	3.2	GAGE
104	Sealant above noted surface, max.		0.40%	3.2	GAGE
105	Sealant not present 360°		0.40%	3.2	VISUAL
106	Assembly damaged or distorted		0.40%	3.2	VISUAL
107	Bridgewire continuity - resistance		100%	3.2	4.5.7
108	Moisture content of barium nitrate		-	3.3.1	4.5.2
109	Moisture content of lead azide		-	3.3.2	4.5.3
110	Insulation resistance test, shorted electrodes to cup		0.40%	3.2	4.5.8
	Sensitivity and Output Test (70° + 10°F)	125			4.5.4/4.4.5.1
111	No-fire test	(a)	125-1-2	3.4.1.1	4.5.4.1
112	All-fire output test, sleeve enlargement		125-1-2	3.4.1.2	4.5.4.2
113	All-fire output test, Dent in place		125-1-2	3.4.1.2	4.5.4.2

NOTES:
(a) 125 Samples from No-fire test to be used to perform All-fire tests below.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.8	Primer, Battery, Electric				9292624
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Environmental Tests - Insulation resistance, thermal shock, vibration and function	25(a)			4.5.5
114	Insulation resistance		25-0-1	3.2	4.5.8
115	No-fire test		25-0-1	3.4.1.1	4.5.5.1
116	All-fire output test, sleeve enlargement		25-0-1	3.4.1.2	4.5.5.1
117	All-fire output test, dent in plate		25-0-1	3.4.1.2	4.5.5.1
	Waterproofness test	50			4.5.5.2
118	All-fire output test, sleeve enlargement		50-0-1	3.4.1.2	4.5.4.2
119	All-fire output test, dent in plate		50-0-1	3.4.1.2	4.5.4.2
Minor					
201	Length of electrodes to twist, min.		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

(a) 25 Samples conditioned as indicated shall be subjected to each of tests below.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Packaging, Packing, Marking		9292624
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Partioning or cushioning improper		0.40%	3.2	5.1-VISUAL
102	Marking of inner packaging missing or improper		0.40%	3.2	5.1-VISUAL
103	Outer box damaged or not fully secured		0.40%	3.2	5.2-VISUAL
104	Marking of outer box missing, illegible or improper		0.40%	3.2	5.2-VISUAL
Minor					
201	Evidence of poor workmanship		0.65%		VISUAL

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the “Paragraph Reference/Inspection Method” column of the First Article or Quality Conformance Inspection tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.5 Function test equipment. The test fixture and procedure for subjecting the electric battery primer to no-fire and all-fire output tests, as evidenced by diameter expansion of witness sleeve and indenting of witness plate shall be that described in 4.4.5.1.

4.4.5.1 Battery primer function fixture. The test fixture and procedure shall be as specified on Dwg.’s 9275627, 9275629, 9275630, 9275631, 9275632, and 9275633 with the following modifications:

a. The diameter of the point or indenting diameter of Support Pin, Dwg. 9275663, shall be changed from ,045” - .003” to 0.55” ± .001”.

b. The material for the Witness Sleeve, Dwg. 9275630, shall be copper-zinc lead alloy rod, alloy 356, temper-half hard, to meet a hardness equivalent to Rockwell A40-55 when measured on the outside diameter.

c. Primer, Dwg. 9275621, is replaced by Primer, Dwg. 9292624.

4.5 Test methods and procedures.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspection and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

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4.5.2 Moisture content of barium nitrate. The moisture content of barium nitrate shall be determined at the time of loading. The method shall be in accordance with MIL-STD-650, Method 101.5. The sample shall be selected from that quantity to be used to produce a primer lot and shall meet requirements of the applicable drawing. If the moisture content fails to comply with specified requirement of 3.3.1, the batch shall be rejected.

4.5.3 Moisture content of lead azide. The moisture content of lead azide shall be determined as specified in 4.5.2, except that a two gram sample shall be used and the dish covered with a sheet of polytetrafluoroethylene. The sample shall be selected from that quantity to be used to produce a primer lot and shall meet the requirements of the applicable drawing. If the moistdure content fails to comply with specified requirement of 3.3.2, the batch shall be rejected.

4.5.4 Functioning tests - Sensitivity and output. The battery primer samples selected from each inspection lot shall be assembled to the applicable test fixture (see 4.4.5.1) and subjected to a no-fire test followed by an all-fire output test.

4.5.4.1 No-fire test (70° ± 10°F). The battery primers shall be conditioned at 70° ± 10°F for two (2) hours minimum (min.) prior to function test. The battery primers shall be assembled to the applicable test fixture (see 4.4.5.1) and subjected to a constant 10 ma current for 10 seconds min. to determine conformance with 3.4.1.1. Any primer which fails to comply with the applicable requirement shall be classed defective. (Non-destructive test).

4.5.4.2 All-fire output test (70° ± 10°F). Following the No-fire test of 4.5.4.1, the battery primers shall be fired by applying the energy from a 100 microfarad capacitor charged to 1.6 volts max. Observation shall be made for failure to comply with requirements specified in 3.4.1.2. If any battery primer fails to meet the output requirements on the first attempt, it shall be classed defective. (Destructive test).

4.5.5 Environmental tests.

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4.5.5.1 Insulation resistance, thermal shock, vibration and function. The battery primer sample shall be subjected to the required tests in the following sequence:

a. Measure insulation resistance between the two shunted electrodes and cup in accordance with 4.5.8.

b. Condition for two (2) hours min. at minus 60° - 5°F.

c. Remove from cold chamber and within 20 seconds transfer to a hot chamber at 155° + 5°F.

d. Condition for two (2) hours min. at 155 + 5°F and 20% max. relative humidity.

e. Remove from hot chamber.

f. Vibrate for 15 minutes, minimum in a sweep covering a frequence range of 25-500-25 cycles per second, parallel to the transverse orthogonal axis, with base charge down and a 2 ± 0.2 G's peak amplitude.

g. Measure resistance between shunted electrodes and cup in accordance with 4.5.8.

h. The primers shall be assembled to test fixture (see 4.4.5.1) and functioned as specified in 4.5.4.1 and 4.5.4.2. Observation shall be made to determine compliance with requirements of 3.4.1 and 3.5.1. Any primer which fails to comply with the applicable requirement shall be classed defective. (Destructive test).

4.5.5.2 Waterproofness. Fifty (50) battery primers from each lot shall be subjected to this test. The battery primer shall be immersed in water to a min. depth of two (2) inches for 48 hours, min. The required water temperature of 70° ± 5°F shall be maintained for the immersion period. At the end of the immersion time, the battery primer shall be removed from the water, assembled into a test fixture (see 4.4.5.1) and fired by applying the energy from a 100 microfarad capacitor charged to 1.6 volts, max. Observation shall be made to determine compliance with the requirements of 3.4.1.2. If any battery primer fails to meet the requirements on the first attempt it shall be classed defective. If one or more defective occur, the lot shall be rejected. (Destructive test)

NOTE: An alternative method for testing seal integrity may be submitted by the contractor.

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4.5.6 Pull test of electrodes of header (Dwg. 9292689). The header shall be assembled in an approved fixture and the procedure of MIL-STD-202, Method 211, Test Condition A shall be used, with the exception that the force specified on the drawing shall be applied to each electrode for 45 seconds. The force shall be a gradual, static axial load. Any header in which either electrode moves or loosens shall be classed defective. This test is a destructive test.

4.5.7 Bridgewire continuity resistance of battery primer. Prior to shorting (twisting) the electrodes, the battery primer shall be connected to approved equipment and the bridgewire (electrode to electrode) resistance measured to determine compliance with the applicable requirement specified on Dwg. 9292624. Any primer which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The measurement shall be made with a meter capable of measurement to the nearest 0.01 ohm. The current applied shall be limited to one (1) milliampere, max. (Non-destructive test).

4.5.8 Insulation resistance of battery primer. The isulation resistance between the shorted (twisted) electrodes and the battery primer cup shall be determined using the procedure specified in Method 302 of MIL-STD-202. The specified voltage on drawing shall be applied for 2 seconds min. Observation shall be made to determine compliance with the applicable requirement specified on Dwg. 9292624. Any primer which fails to comply with the applicable requirement shall be classed defective. (Non-destructive test).

5. PACKAGING AND PACKING

5.1 Packaging. The electric battery primer shall be packaged in accordance with level B of Spec MIL-P-14232. The primers shall be partitioned and cushioned as required.

5.2 Packing. The electric battery primer shall be packed in accordance with Level B of Spec MIL-P-14232 and Code of Federal Regulations, Title 49 with the following mandatory requirement. The outer packing box shall be wood conforming to class 1, to style 4, Spec PP-B-621 or fiberboard conforming to Spec 12H, 23F or 23H of CFR, Title 49.

5.3 Marking. The marking of the outer box shall be in accordance with MIL-STD-129 and DOT marking per CFR, Title 49.

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5.4 Shipping. When primers from more than one lot arc shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on mines: XM74 AP, XM75 AT/AV, BLU-91/B AT/AV and BLU-92/B AP.

6.2 Ordering data. See MIL-A-48078(PA).

6.3 Submissions of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required, to COMMANDER: ARRADCOM, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Distribution of ammunition data cards. Distribution of data cards shall include the following: COMMANDER, ARRADCOM, ATTN: DRDAR-QAA-Q, Dover, New Jersey 07801.

6.5 Check test for possible deterioration. The following test is designed to prevent primers that have deteriorated from being assembled into major ammunition items. The test shall be performed on primers from each lot which have been stored under normal conditions for more than two (2) years or under adverse conditions for any period of time. If the specification for the ammunition item into which primers are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of check tests for possible deterioration. Primers shall be selected from each lot for this test as per 6.5.1.

6.5.1 Sensitivity and output. Two hundred (200) battery primers shall be selected from each lot for these tests. The test shall be performed as specified in 4.5.4 using equipment in accordance with 4.4.5. Any primer that fails to comply with the requirements of 3.4.1 shall be classed defective. If three (3) or more primers are classed defective, the lot shall be rejected and not used in the loading operation. (Destructive test).

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6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U. S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARRADCOM.

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Preparing Activity:
Army-AR

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